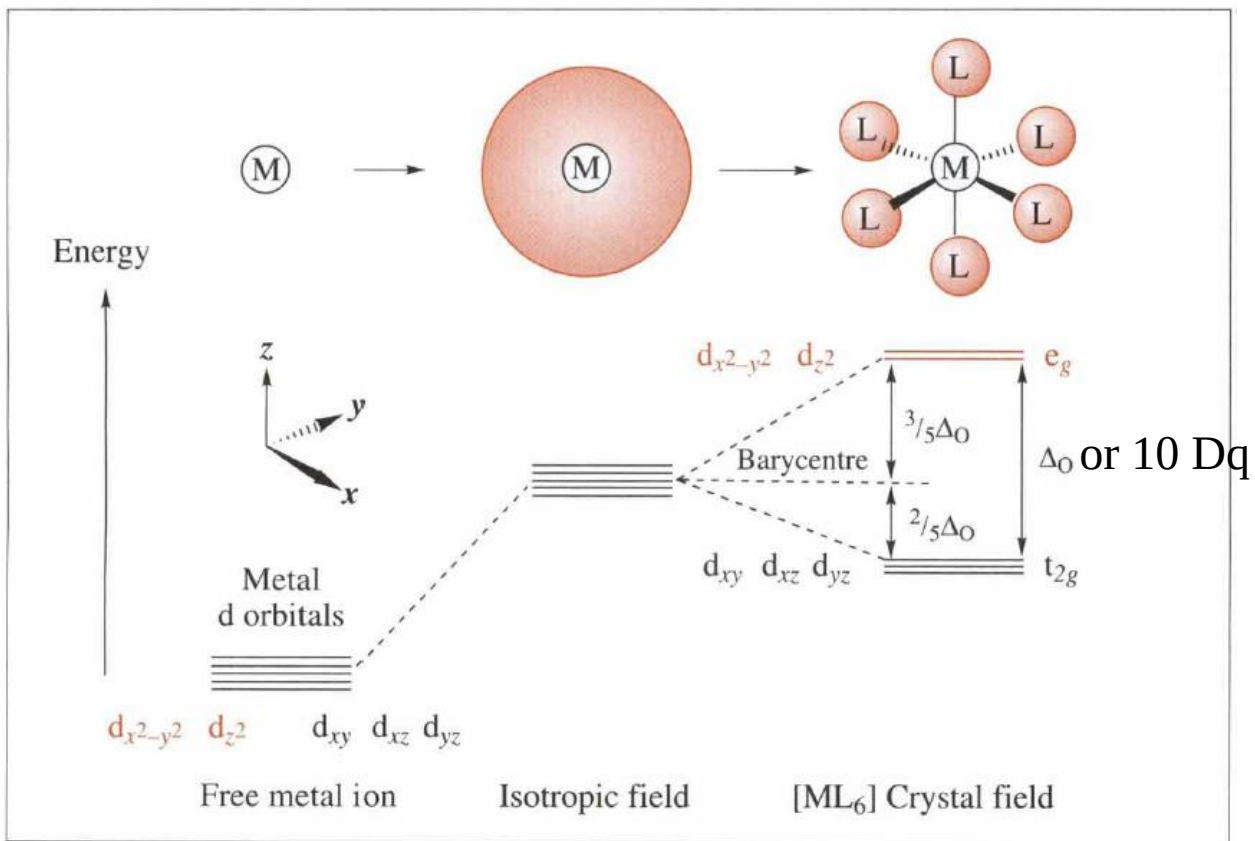
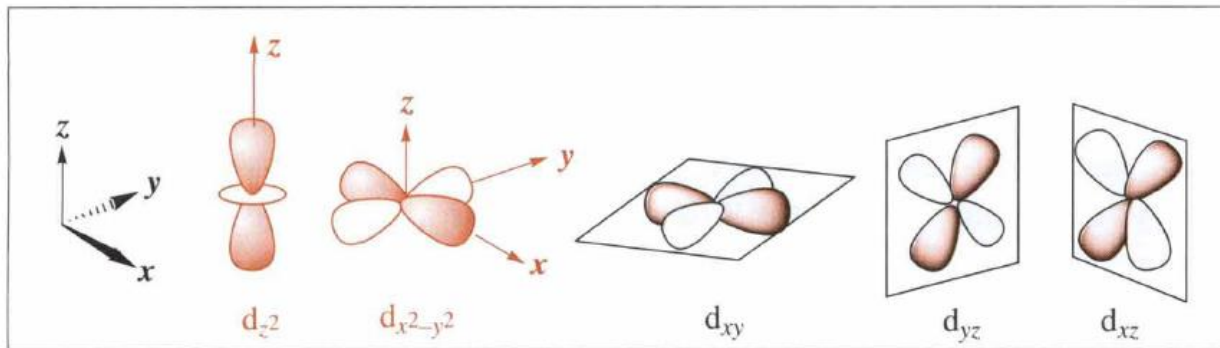
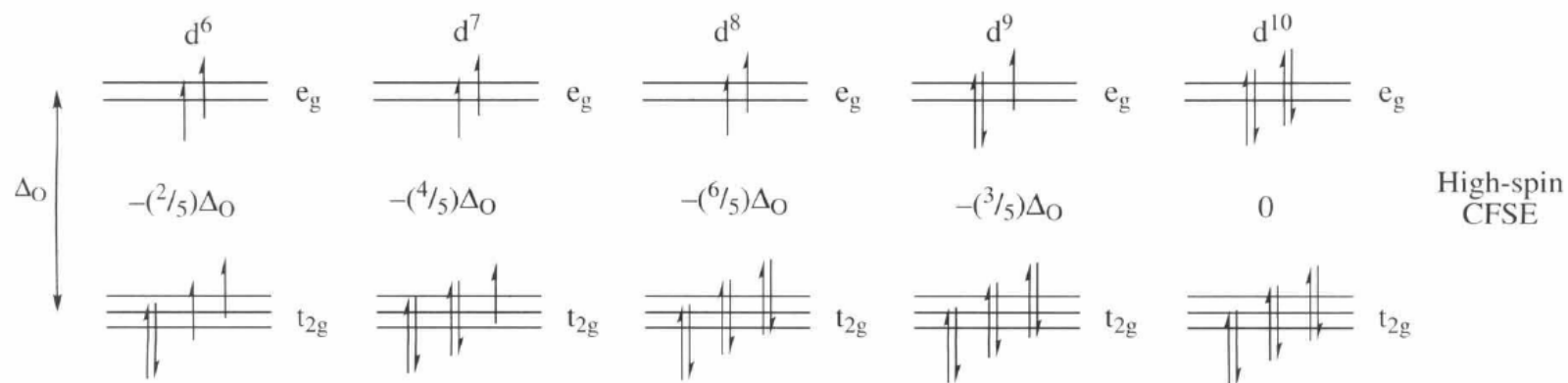
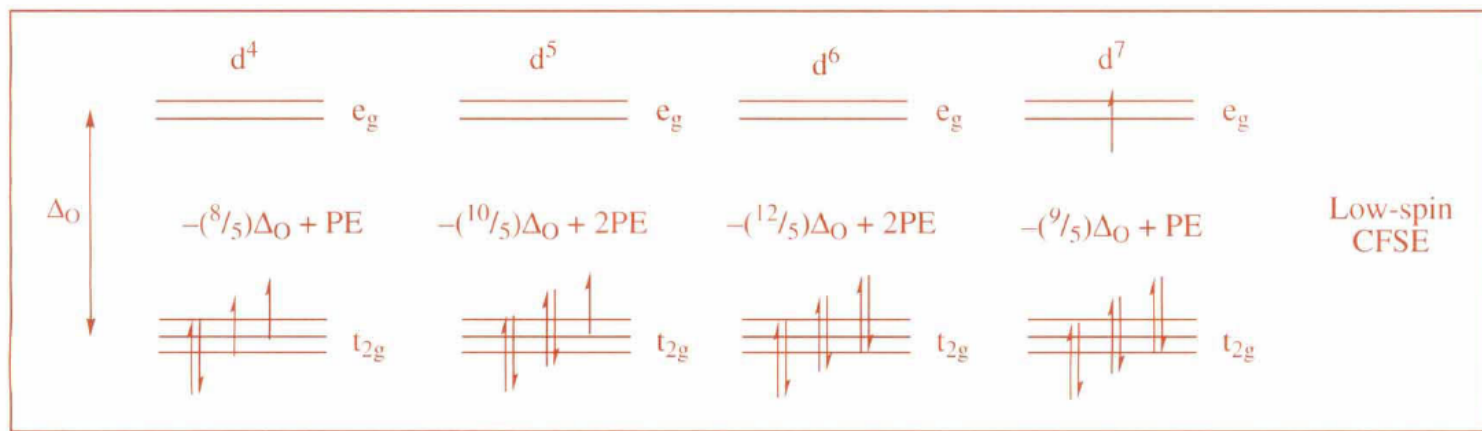
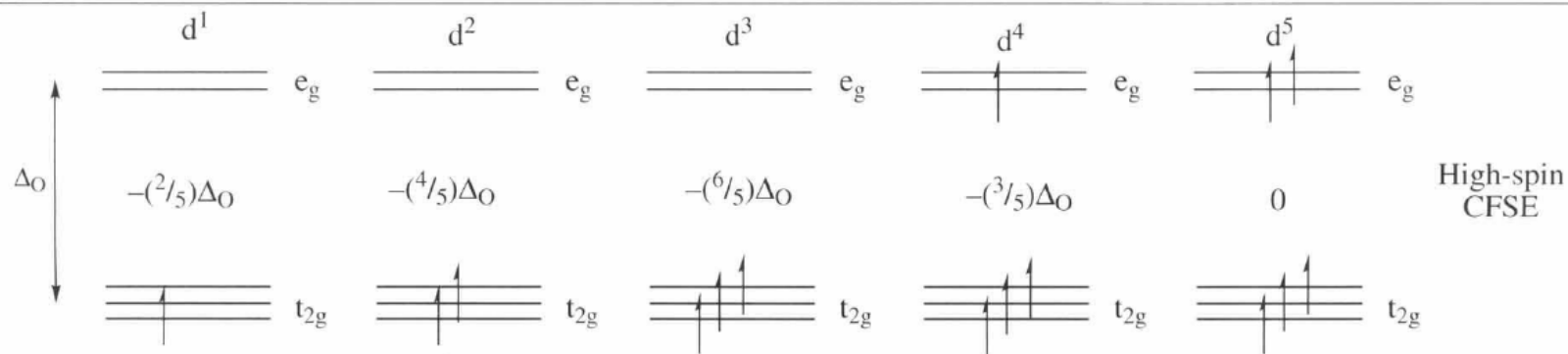
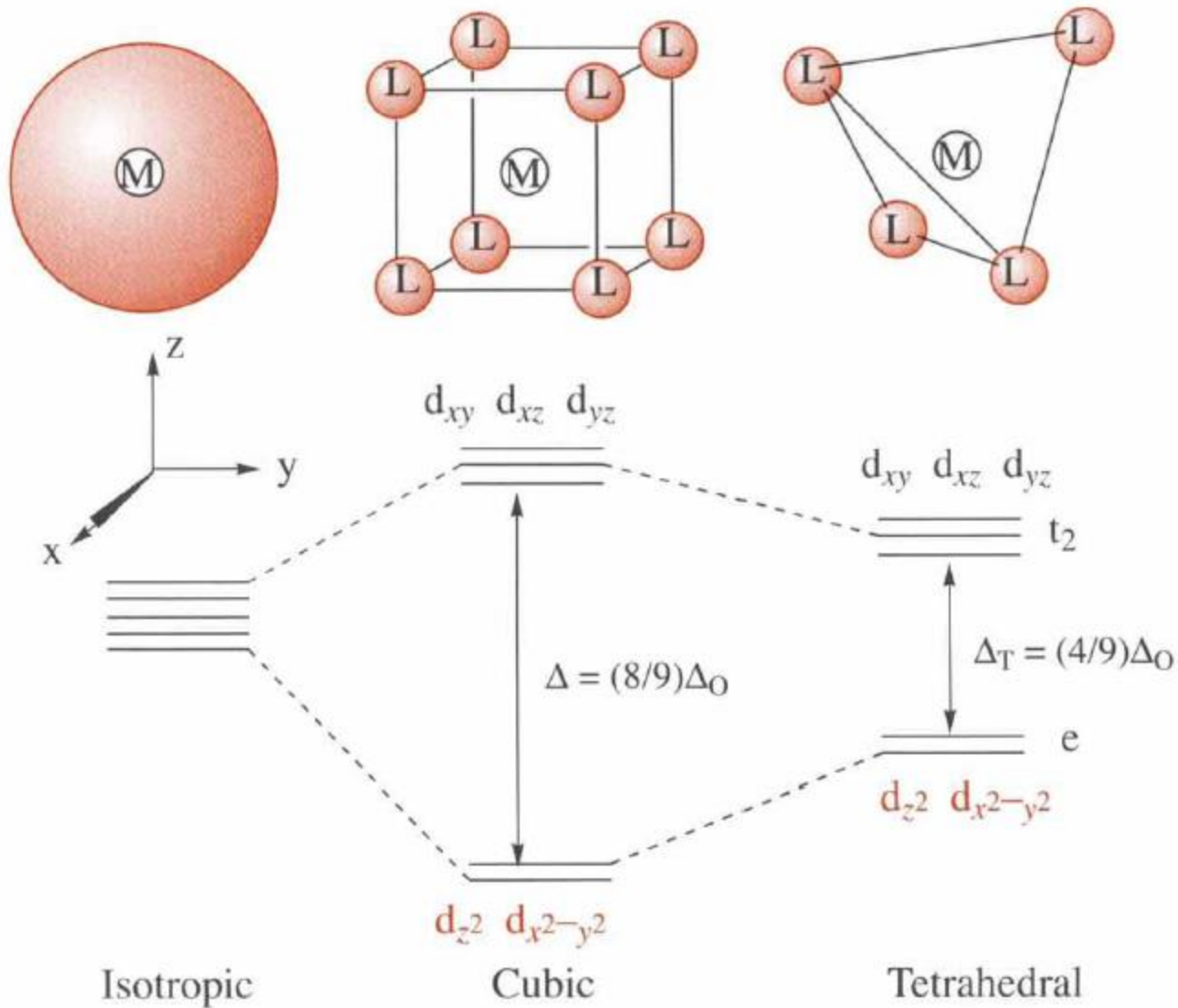


Crystal Field Theory and Ligand Field Theory

Crystal Field Theory (1930)







Jahn-Teller Effect

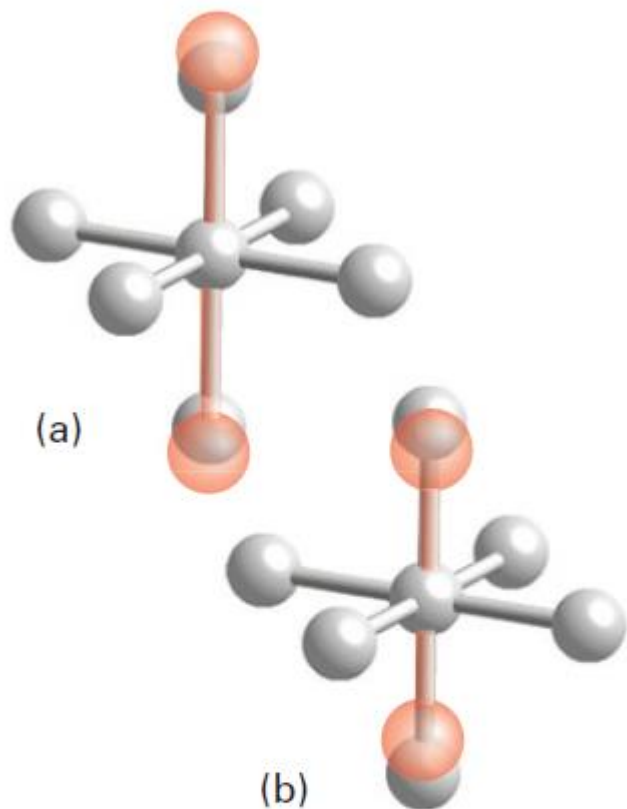


Figure 20.11 (a) A tetragonally distorted complex where two of the ligands have moved further away from the central ion. (b) A tetragonally distorted complex where two of the ligands have moved closer towards the central ion.

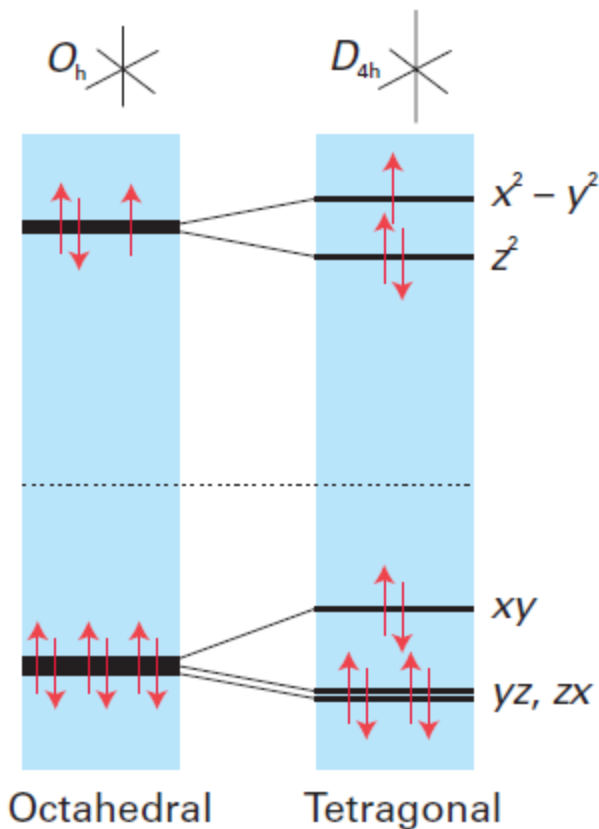
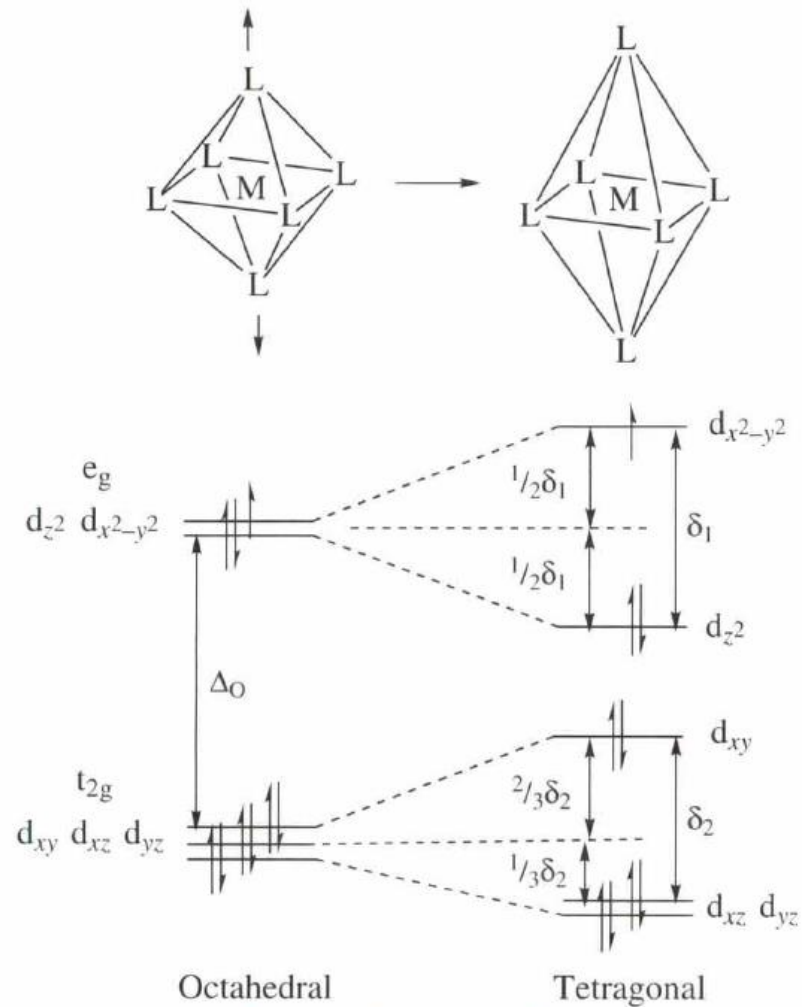


Figure 20.12 The effect of tetragonal distortions (compression along x and y and extension along z) on the energies of d orbitals. The electron occupation is for a d^9 complex.

Show degeneracy

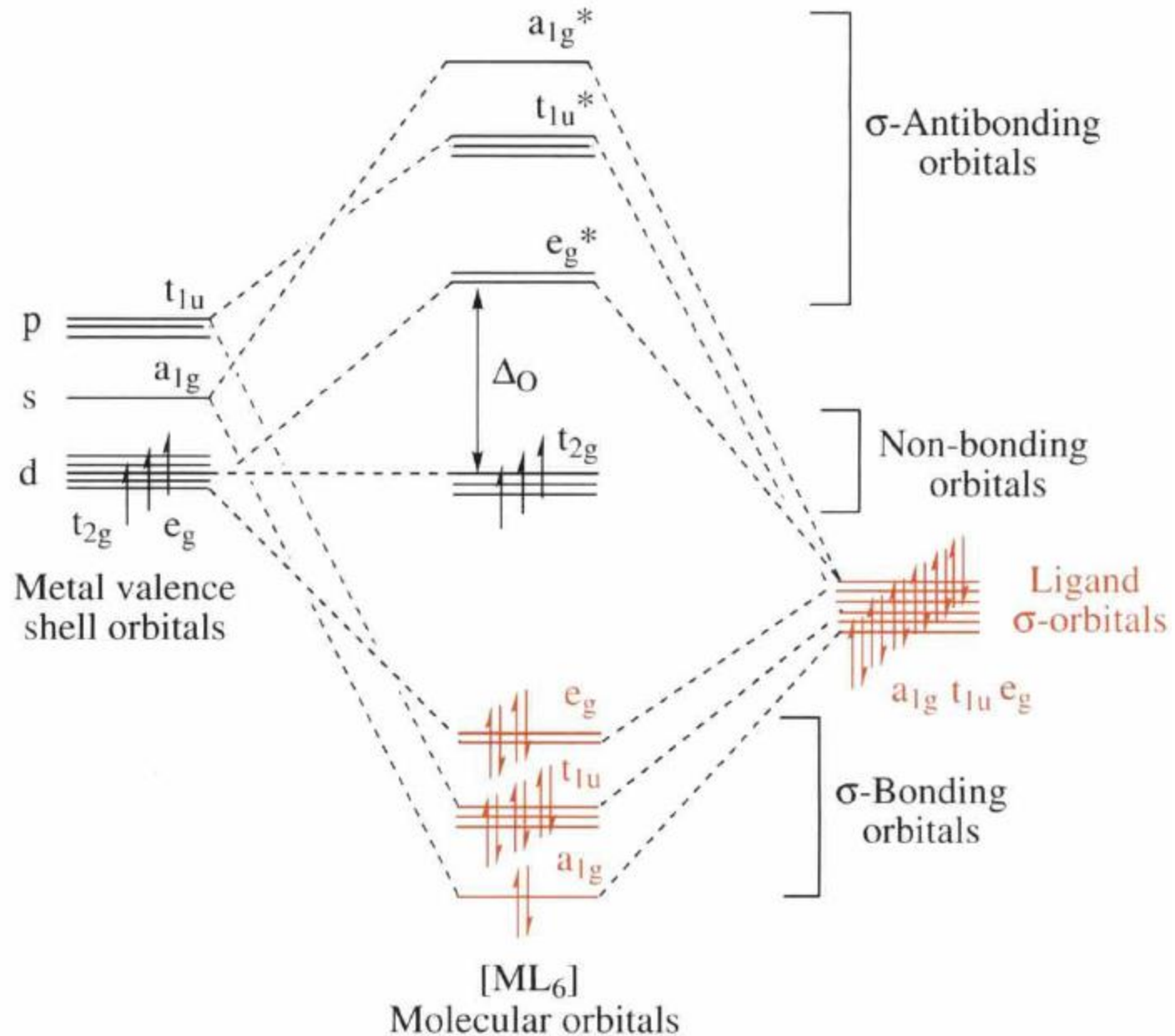
Jahn-Teller Effect

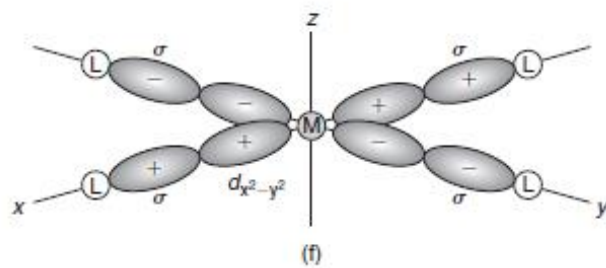
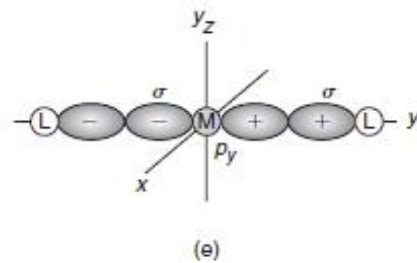
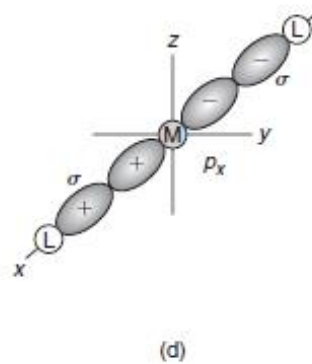
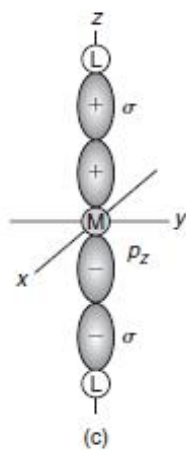
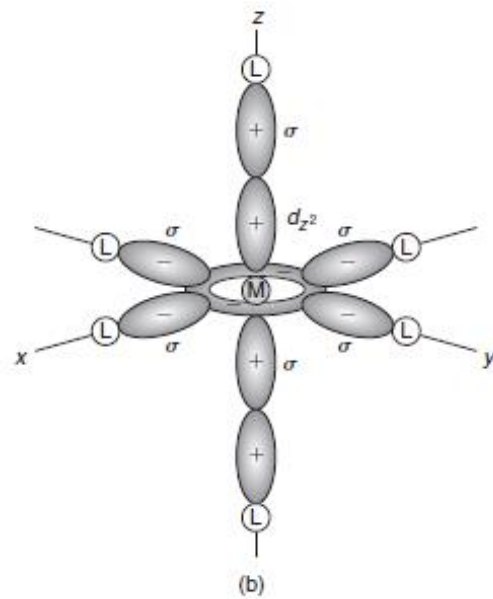
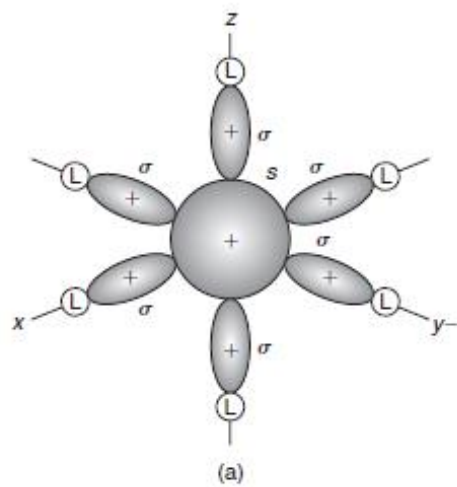


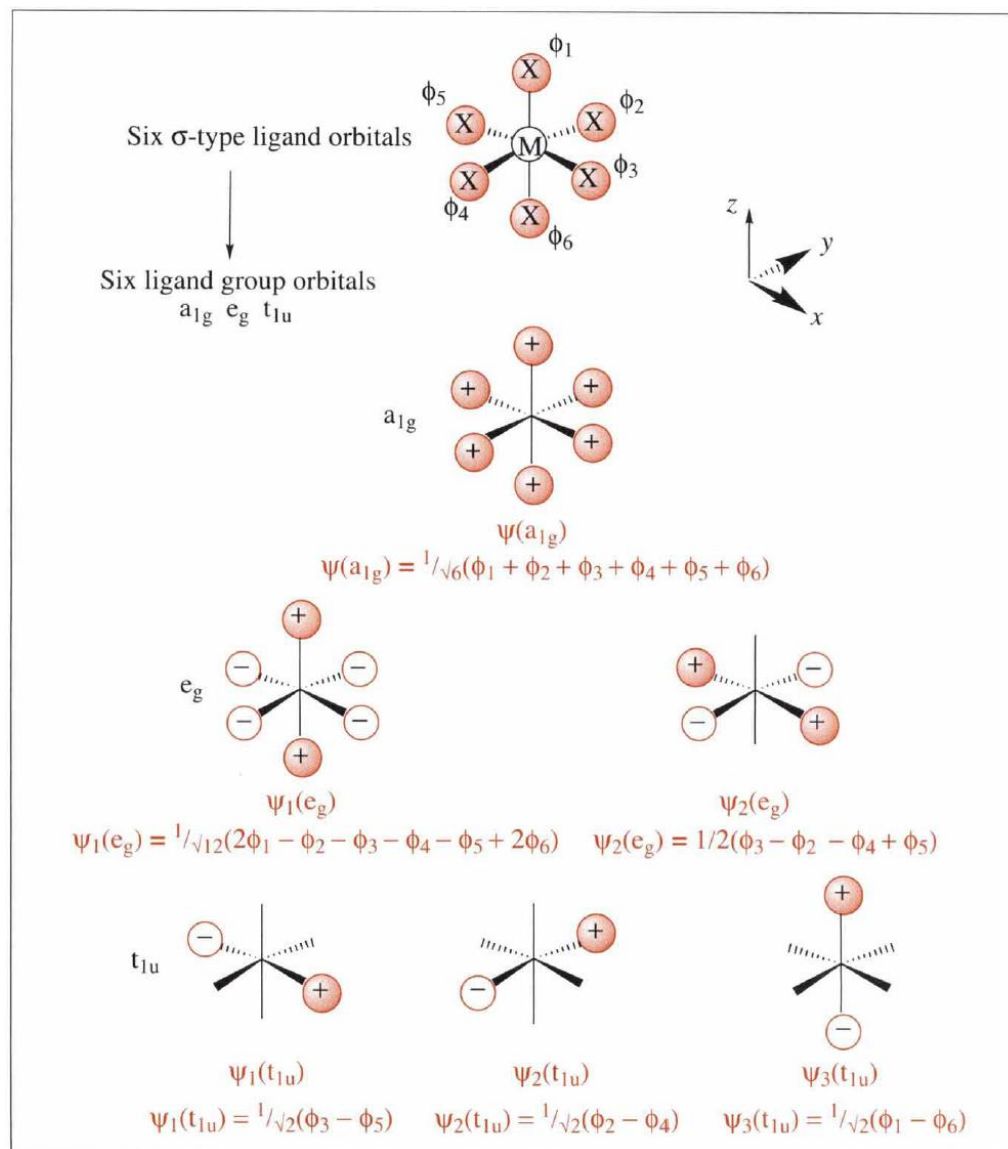
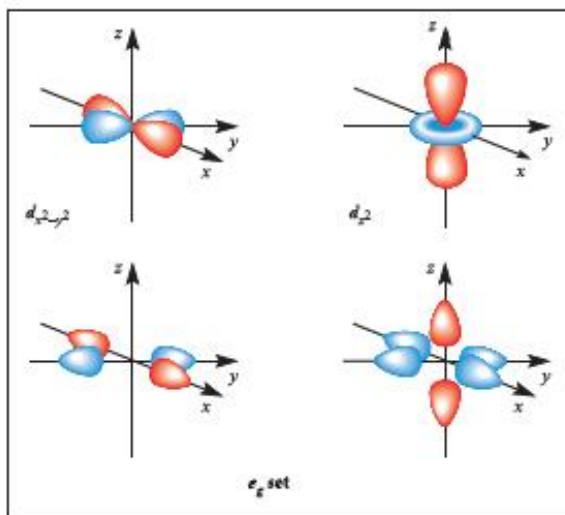
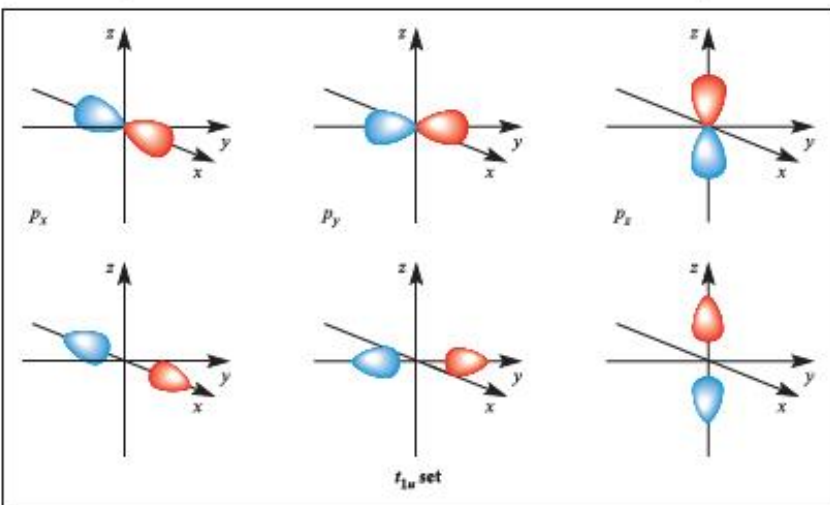
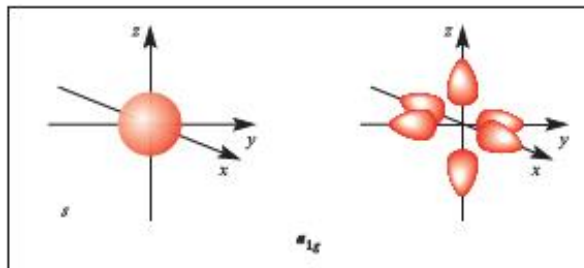
$$\text{Octahedral CFSE} = (3 \times \frac{3}{5} - 6 \times \frac{2}{5})\Delta_O = -(\frac{3}{5})\Delta_O$$

$$\begin{aligned} \text{Tetragonal CFSE} &= -(\frac{3}{5})\Delta_O + (2 \times \frac{2}{3})\delta_2 - (4 \times \frac{1}{3})\delta_2 + (\frac{1}{2})\delta_1 - (2 \times \frac{1}{2})\delta_1 \\ &= -(\frac{3}{5})\Delta_O - (\frac{1}{2})\delta_1 \end{aligned}$$

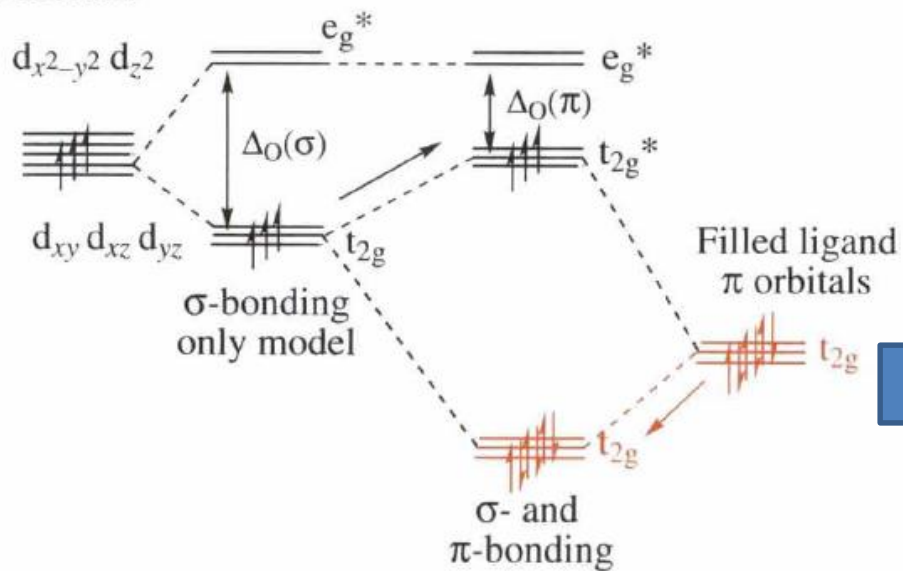
MO Theory and Ligand Field Theory





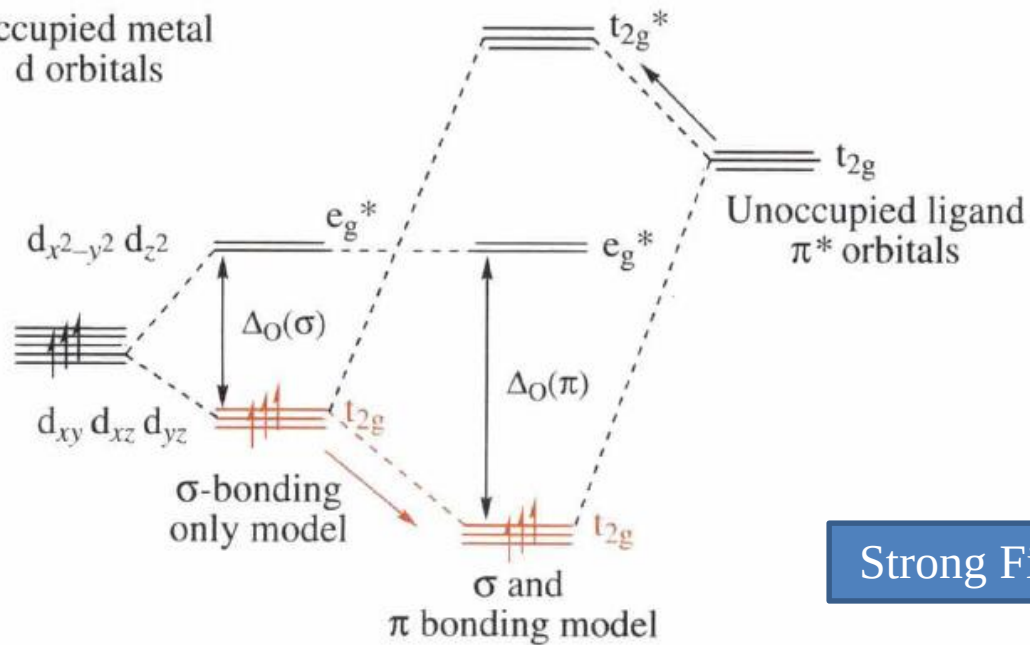


(a) Occupied metal d orbitals

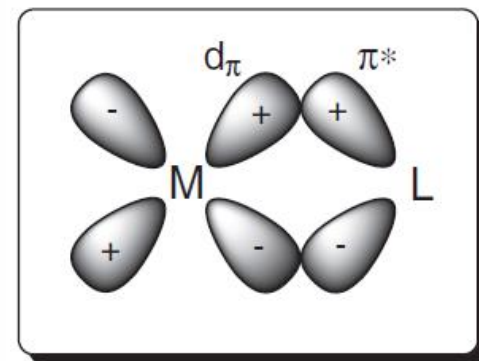


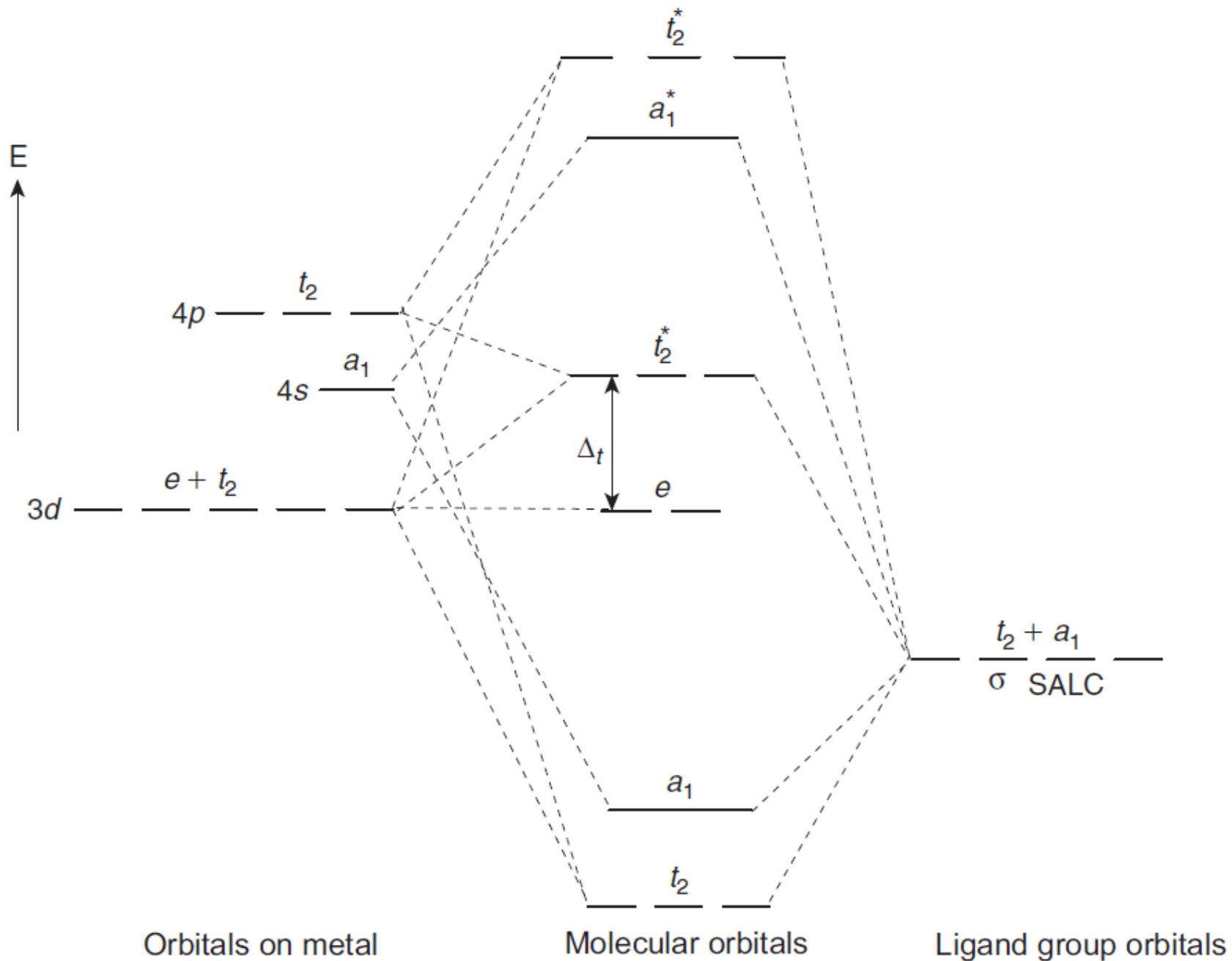
Weak Field Ligand

(b) Occupied metal d orbitals



Strong Field Ligand





Russell-Saunders Ground State Term

TERM STATE SYMBOLS

Note the pattern obtained for term state symbol values upon incrementing the number of d electrons:

d^n	d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
term symbol	2D	3F	4F	5D	6S	5D	4F	3F	2D	1S

Orgel Diagram

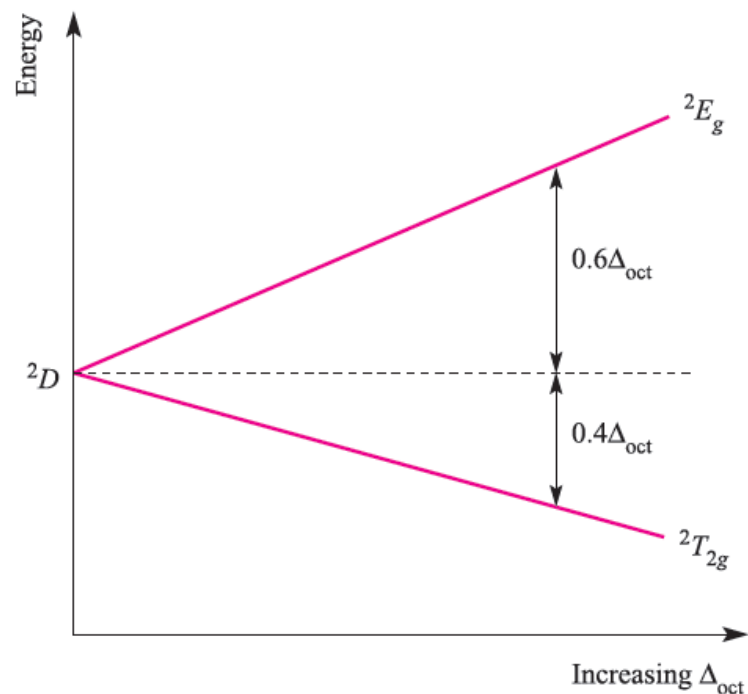
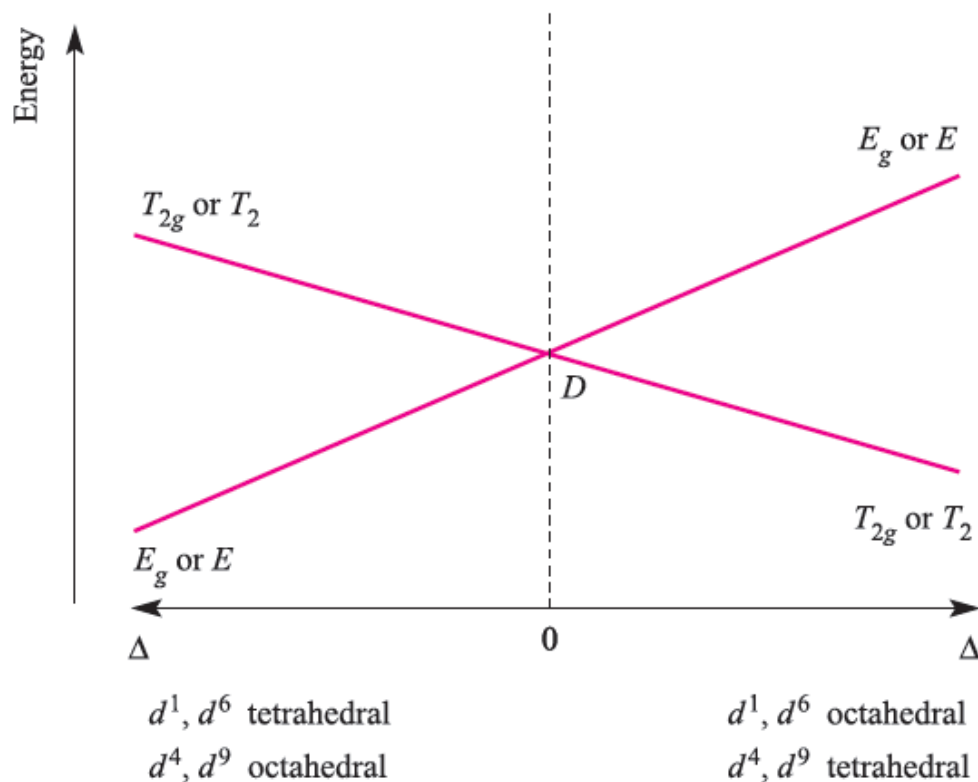


Fig. 20.17 Orgel diagram for d^1, d^4 (high-spin), d^6 (high-spin) and d^9 ions in octahedral (for which T_{2g} and E_g labels are relevant) and tetrahedral (E and T_2 labels) fields. In

Russell-Saunders Ground State Term

TERM STATE SYMBOLS

values upon incrementing the number of d electrons:

d^n	d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
term symbol	2D	3F	4F	5D	6S	5D	4F	3F	2D	1S

Orgel Diagram

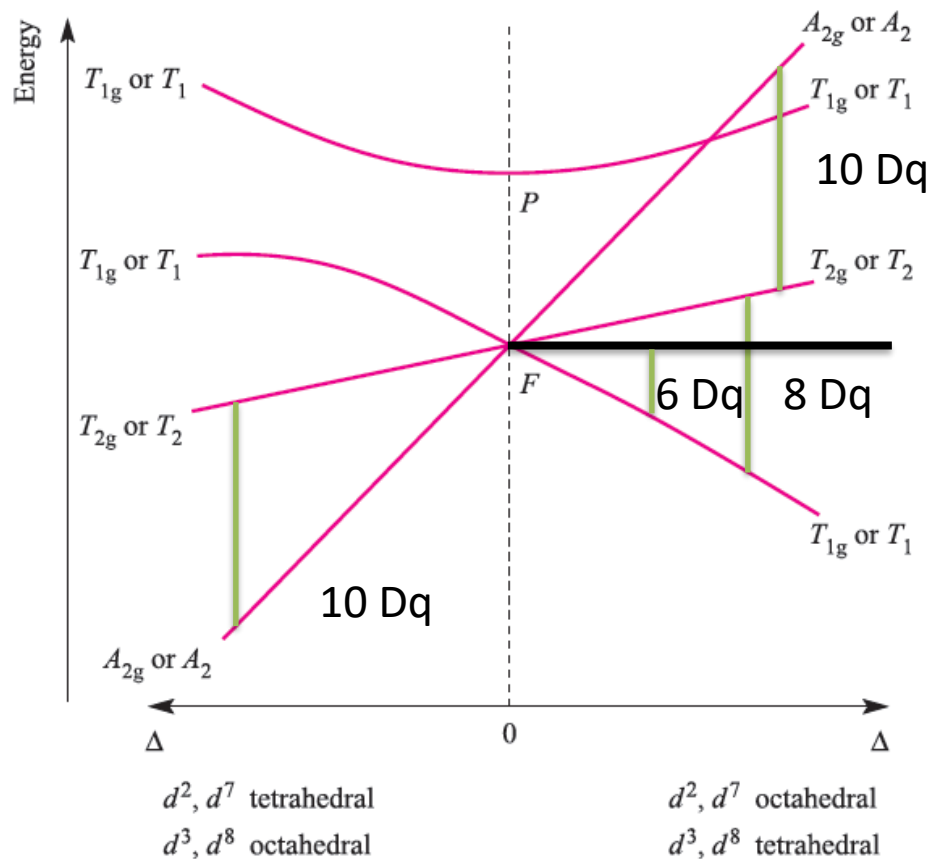


Fig. 20.18 Orgel diagram for d^2 , d^3 , d^7 and d^8 ions (high-spin) in octahedral (for which T_{1g} , T_{2g} and A_{2g} labels are relevant) and tetrahedral (T_1 , T_2 and A_2 labels) fields. Multiplicities are not stated because they depend on the d^n configuration, e.g. for the octahedral d^2 ion, ${}^3T_{1g}$, ${}^3T_{2g}$ and ${}^3A_{2g}$ labels are appropriate.



Table 7.3 Ligand field terms for an octahedral complex Mulliken Symbol

Free ion term	Number of states (2L+1)	Octahedral field terms
S (L = 0)	1	A_{1g}
P (L = 1)	3	T_{1g}
→ D (L = 2)	5	T_{2g}, E_g
→ F (L = 3)	7	T_{1g}, T_{2g}, A_{2g}
G (L = 4)	9	$A_{1g}, E_g, T_{1g}, T_{2g}$

Box 7.3 Hund's Rules

The energies of the various Russell–Saunders states of an atom or ion are not the same, and Hund's first and second rules can be used to decide which is the ground state:

- (i) The lowest energy term (ground state) has the largest spin multiplicity ($2S + 1$).
- (ii) If two terms have the same spin multiplicity, that with the larger value of L is of lowest energy.

These rules are usually accurate in defining the ground state but do not always predict the correct order for the higher energy terms, and for a d^2 ion the energy order of states is, starting from the ground state, ${}^3F < {}^1D < {}^3P < {}^1G < {}^1S$.

Box 7.1 Selection Rules

Two important selection rules govern which electronic transitions of a metal complex may be observed. The first is the **spin selection rule**, which requires that there must be no change in the total spin quantum number S during the transition. As examples, singlet-to-singlet or triplet-to-triplet transitions are allowed, but triplet-to-singlet transitions are forbidden. In practice, spin-forbidden transitions may be observed, but will be very weak having typical molar absorptivities, ϵ , of $<1 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$. A further selection rule, known as the **Laporte rule**, also applies to d–d or f–f transitions. This requires that, for ions in a centrosymmetric environment such as an octahedral field, the parity must change, $g \rightarrow u$ or $u \rightarrow g$, as a result of the electronic transition. In an octahedral complex the d orbitals are all of g parity, so that d–d transitions are Laporte forbidden. In practice, vibrations occurring within the complex can reduce its symmetry on a timescale of $ca. 10^{-13} \text{ s}$. Since the timescale for an electronic transition is much shorter, $ca. 10^{-15} \text{ s}$, they are not totally prevented, and weak absorptions are seen with ϵ values of $ca. 10 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$. In tetrahedral complexes there is no centre of symmetry, so the Laporte rule is relaxed and ϵ values typically fall in the range 100 to $10^3 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$. MLCT and LMCT processes are not confined to metal orbitals alone so are not Laporte forbidden. They are much more intense, with typical ϵ values in the range 10^3 to $10^5 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$.

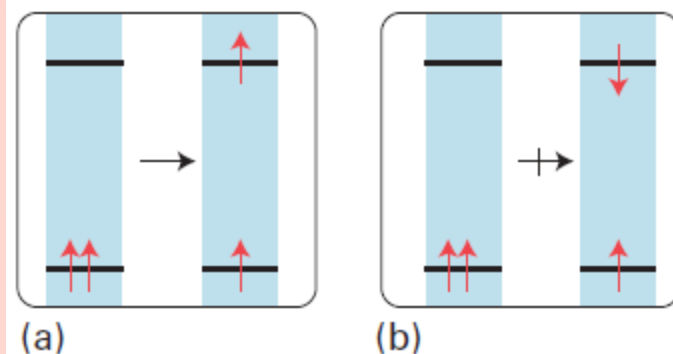
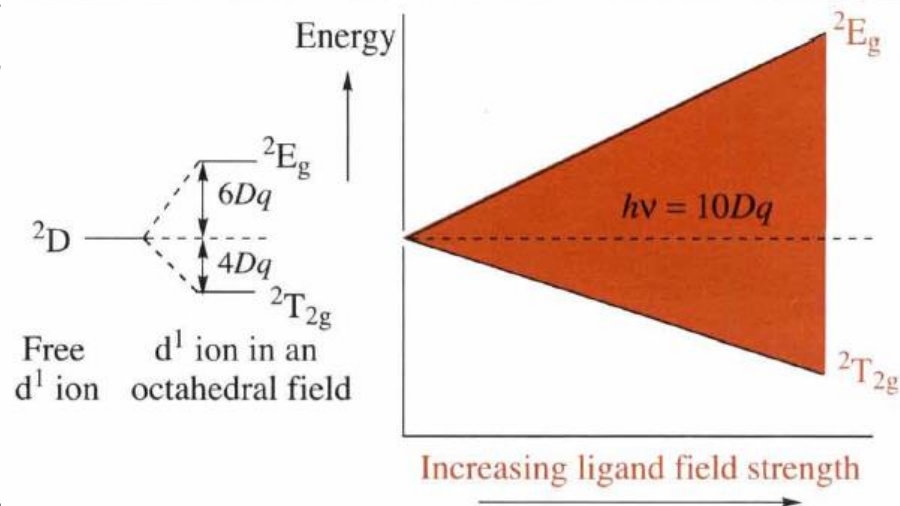
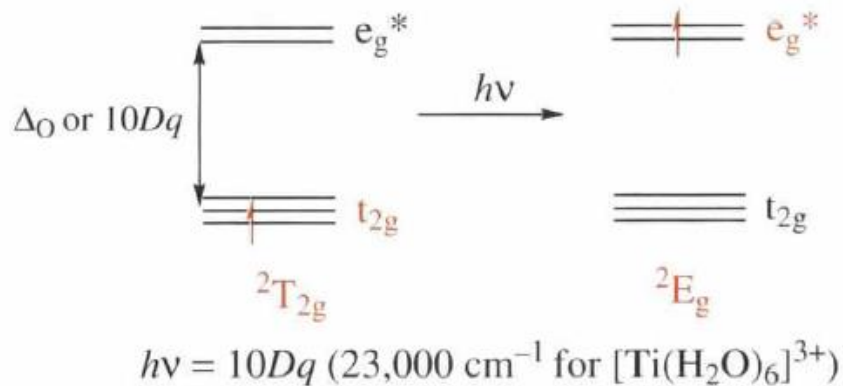
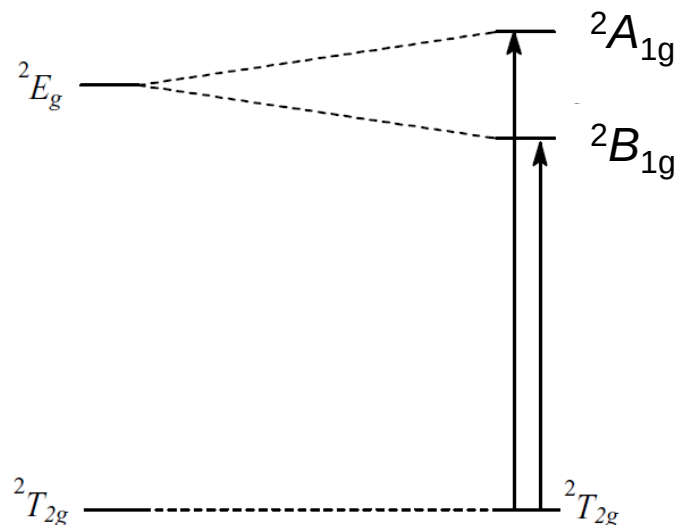
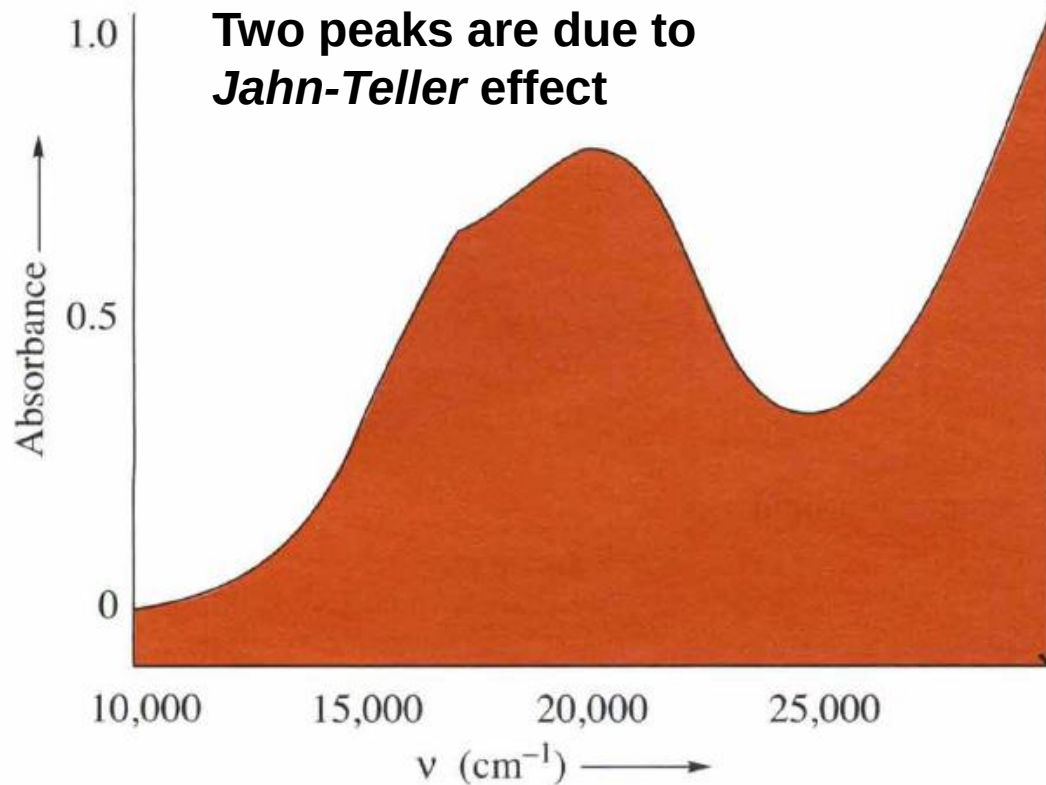


Figure 20.34 (a) A spin-allowed transition does not change the multiplicity. (b) A spin-forbidden transition results in a change in the multiplicity.

Table 20.10 Intensities of spectroscopic bands in 3d complexes

Band type	$\epsilon_{\text{max}} /$ ($\text{dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$)
Spin-forbidden	< 1
Laporte-forbidden d–d	20–100
Laporte-allowed d–d	$c. 250$
Symmetry-allowed (e.g. CT)	1000–50 000

Two peaks are due to *Jahn-Teller effect*



Electronic Spectra of $[M(H_2O)_6]^{n+}$ Complexes

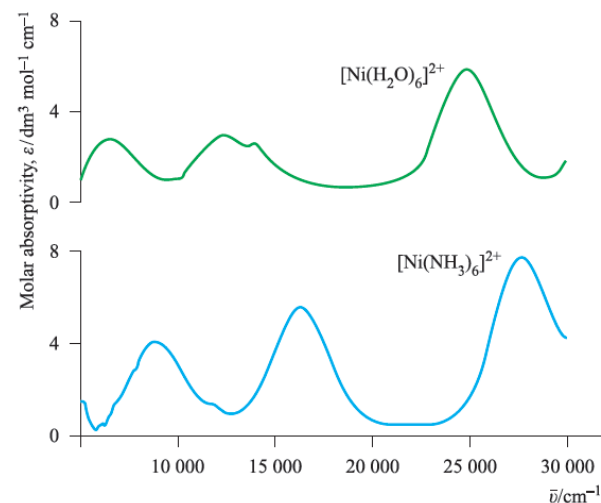
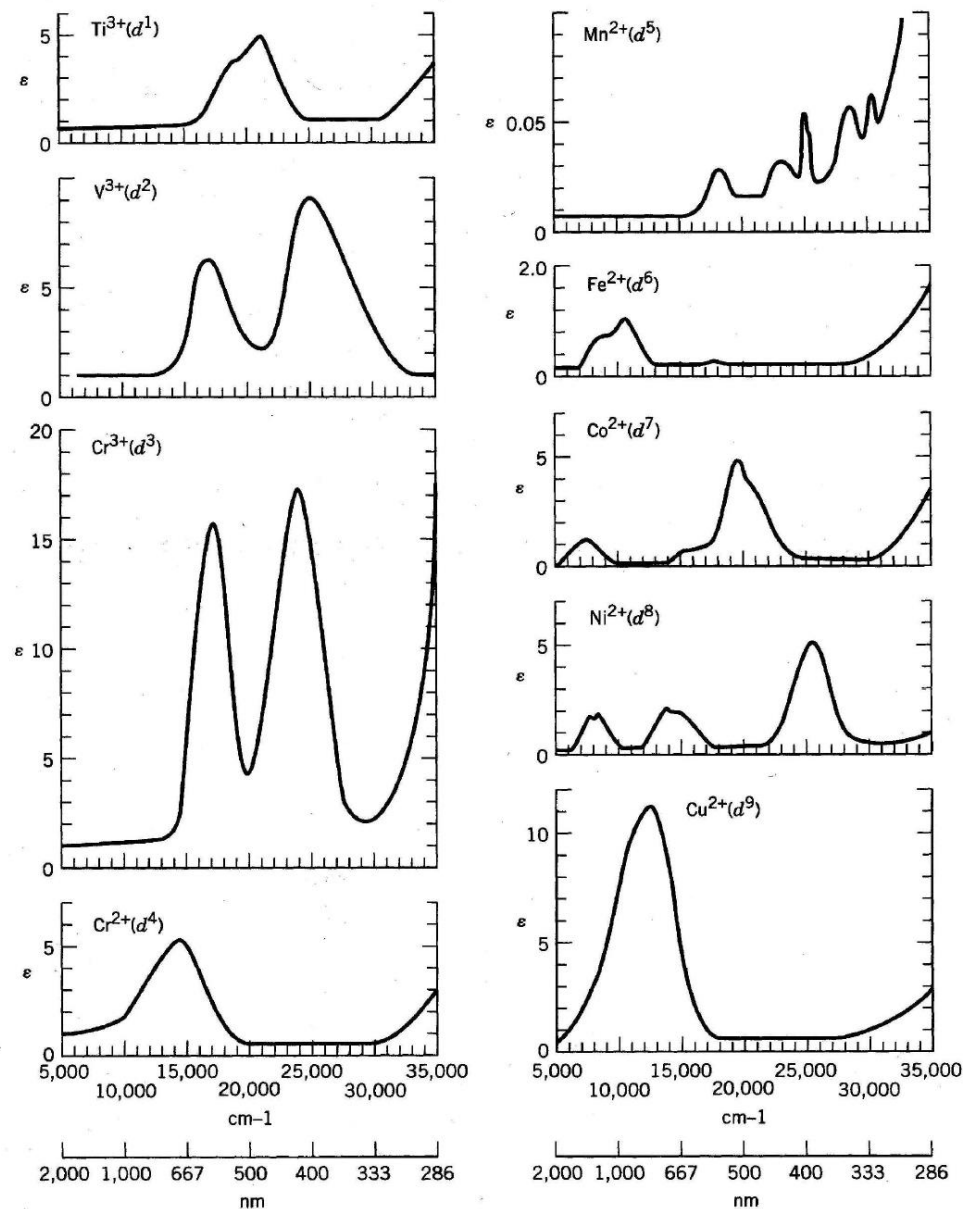
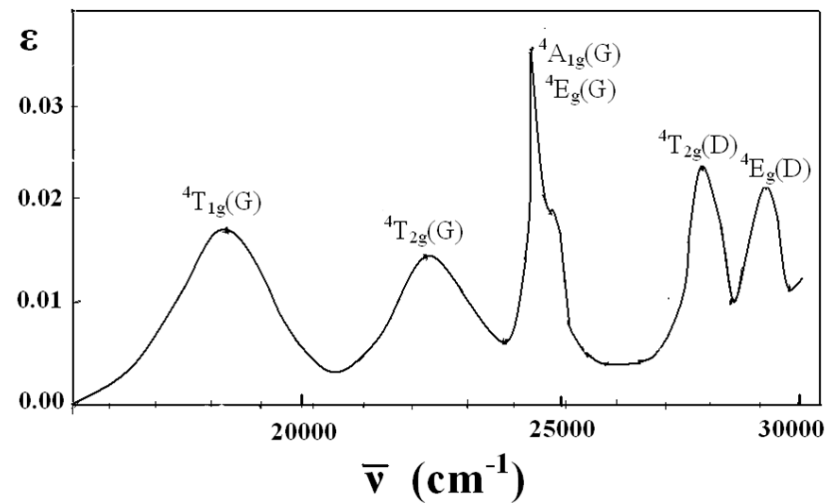
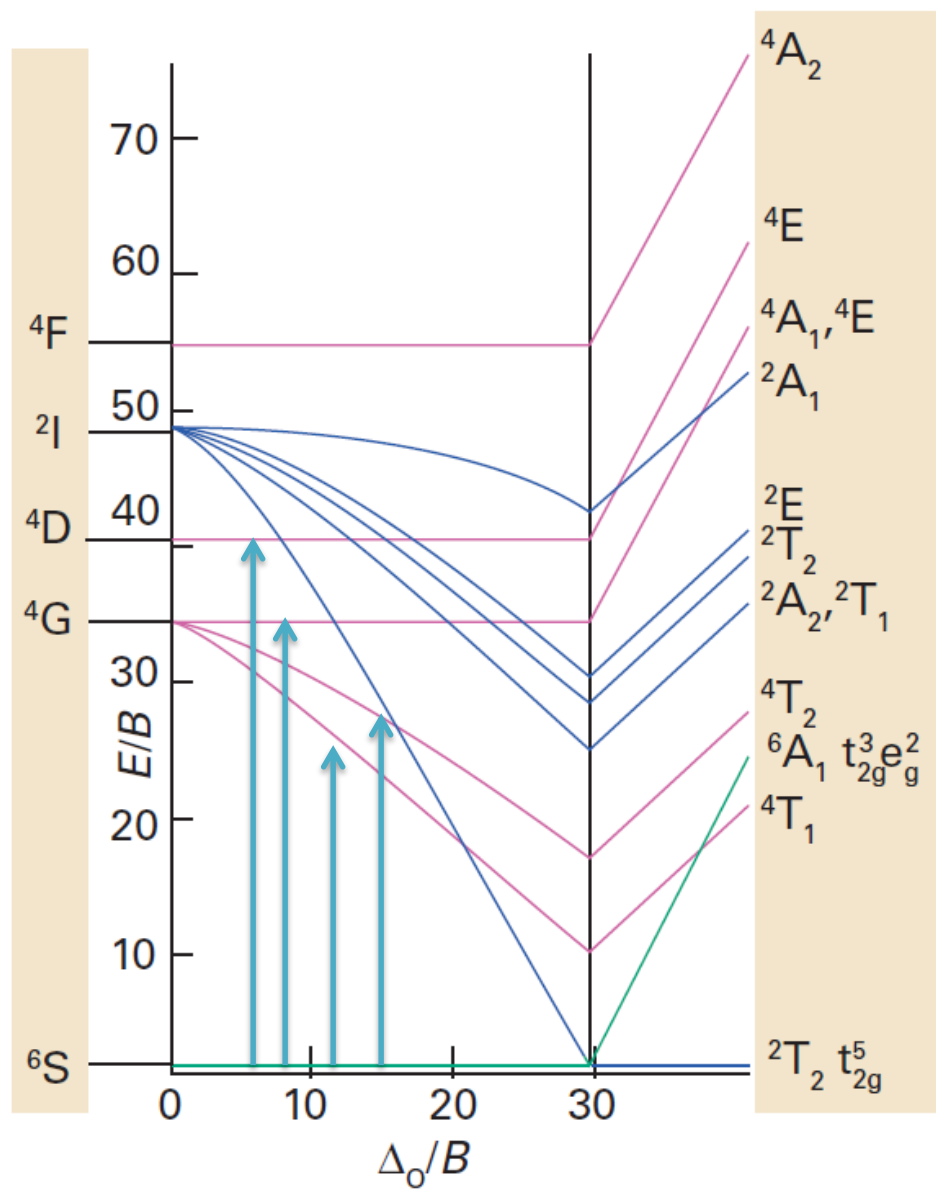


Fig. 20.19 Electronic spectra of $[Ni(H_2O)_6]^{2+}$ ($0.101 \text{ mol dm}^{-3}$) and $[Ni(NH_3)_6]^{2+}$ ($0.315 \text{ mol dm}^{-3}$ in aqueous NH_3 solution) showing three absorption bands.

Explain

d^5 Configuration



Splitting of G and D terms

Tanabe-Sugano Diagram

